

The first equality follows from the law of iterated expectations and (10). The second equality follows from Assumption 7. By the definition of $w_{g,t}$, $\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{g,t} = 1$, hence the third equality. The fourth equality follows from the law of iterated expectations.

Proof of Theorem 2

It follows from the Frisch-Waugh theorem and the definition of $\varepsilon_{fd,g,t}$ that

$$E\left(\widehat{\beta}_{fd} \middle| \mathbf{D}\right) = \frac{\sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t-1} | \mathbf{D}))}{\sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (D_{g,t} - D_{g,t-1})}. \quad (13)$$

Now, by definition of $\varepsilon_{fd,g,t}$ again,

$$\sum_{g=1}^G N_{g,t} \varepsilon_{fd,g,t} = 0 \text{ for all } t \in \{2, \dots, T\}. \quad (14)$$

Then,

$$\begin{aligned} & \sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t-1} | \mathbf{D})) \\ &= \sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (E(Y_{g,t} | \mathbf{D}) - E(Y_{g,t-1} | \mathbf{D}) - E(Y_{1,t} | \mathbf{D}) + E(Y_{1,t-1} | \mathbf{D})) \\ &= \sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (D_{g,t} \tilde{\Delta}_{g,t} - D_{g,t-1} \tilde{\Delta}_{g,t-1} - D_{1,t} \tilde{\Delta}_{1,t} + D_{1,t-1} \tilde{\Delta}_{1,t-1}) \\ &= \sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (D_{g,t} \tilde{\Delta}_{g,t} - D_{g,t-1} \tilde{\Delta}_{g,t-1}) \\ &= \sum_{g,t} (N_{g,t} \varepsilon_{fd,g,t} - N_{g,t+1} \varepsilon_{fd,g,t+1}) D_{g,t} \tilde{\Delta}_{g,t} \\ &= \sum_{(g,t):D_{g,t}=1} N_{g,t} \left(\varepsilon_{fd,g,t} - \frac{N_{g,t+1}}{N_{g,t}} \varepsilon_{fd,g,t+1} \right) \tilde{\Delta}_{g,t}. \end{aligned} \quad (15)$$

The first and third equalities follow from (14). The second equality follows from Lemma 1. The fourth equality follows from a summation by part, and from the fact $\varepsilon_{fd,g,1} = \varepsilon_{fd,g,T+1} = 0$. The fifth equality follows from Assumption 2.

A similar reasoning yields

$$\sum_{(g,t):t \geq 2} N_{g,t} \varepsilon_{fd,g,t} (D_{g,t} - D_{g,t-1}) = \sum_{(g,t):D_{g,t}=1} N_{g,t} \left(\varepsilon_{fd,g,t} - \frac{N_{g,t+1}}{N_{g,t}} \varepsilon_{fd,g,t+1} \right). \quad (16)$$

Combining (13), (15), (16), and the law of iterated expectations yields the result.

Proof of Proposition 2

It follows from the first order conditions attached to Regression 2 and a few lines of algebra that $\varepsilon_{fd,g,t} = D_{g,t} - D_{g,t-1} - D_{.,t} + D_{.,t-1}$. Therefore, under Assumption 6 and if $N_{g,t}$ does not vary across t , one has that for all (g, t) such that $D_{g,t} = 1, 1 \leq t \leq T-1$, $w_{fd,g,t}$ is proportional to $1 - D_{g,t-1} - (2D_{.,t} - D_{.,t-1} - D_{.,t+1})$. $D_{.,t} - D_{.,t-1} \leq 1$, and under Assumption 6 $D_{.,t} - D_{.,t+1} \leq 0$, so $1 - D_{g,t-1} - (2D_{.,t} - D_{.,t-1} - D_{.,t+1})$ can only be strictly negative if $D_{g,t-1} = 1$. Then, for all (g, t) such that $D_{g,t} = 1, 1 \leq t \leq T-1$, $w_{fd,g,t}$ is strictly negative if and only if $D_{g,t-1} = 1$ and $2D_{.,t} - D_{.,t-1} - D_{.,t+1} > 0$.

Similarly, when $t = T$, under the same assumptions as above, one has that for all g such that $D_{g,T} = 1$, $w_{fd,g,T}$ is proportional to $1 - D_{g,T-1} - (D_{.,T} - D_{.,T-1})$. $D_{.,T} - D_{.,T-1} \leq 1$, so $1 - D_{g,T-1} - (D_{.,T} - D_{.,T-1})$ can only be strictly negative if $D_{g,T-1} = 1$. Then, $w_{fd,g,T}$ is strictly negative if and only if $D_{g,T-1} = 1$ and $D_{.,T} - D_{.,T-1} > 0$.

Finally, when $t = 1$, one has that for all g such that $D_{g,1} = 1, D_{g,2} = 1$ under Assumption 6, so $w_{fd,g,1}$ is proportional to $D_{.,2} - D_{.,1}$, which is greater than 0 under Assumption 6.

Proof of Theorem 3

First, by definition of DID_M ,

$$E(\text{DID}_M) = \sum_{t=2}^T E \left(\left(\frac{N_{1,0,t}}{N_S} E(\text{DID}_{+,t} | \mathbf{D}) + \frac{N_{0,1,t}}{N_S} E(\text{DID}_{-,t} | \mathbf{D}) \right) \right). \quad (17)$$

Let t be greater than 2, and let us focus for now on the case where there is at least one g_1 such that $D_{g_1,t-1} = 0$ and $D_{g_1,t} = 1$. Then Assumption 11 ensures that there is a least another group g_2 such that $D_{g_2,t-1} = D_{g_2,t} = 0$. For every g such that $D_{g,t-1} = 0$ and $D_{g,t} = 1$, we have

$$E(Y_{g,t} - Y_{g,t-1} | \mathbf{D}) = E(\Delta_{g,t} | \mathbf{D}) + E(Y_{g,t}(0) - Y_{g,t-1}(0) | \mathbf{D}). \quad (18)$$

Under Assumptions 12, 4, and 5, for all $t \geq 2$, there exists a real number $\psi_{0,t}$ such that for all g

$$E(Y_{g,t}(0) - Y_{g,t-1}(0) | \mathbf{D}) = E(Y_{g,t}(0) - Y_{g,t-1}(0) | \mathbf{D}_g) = E(Y_{g,t}(0) - Y_{g,t-1}(0)) = \psi_{0,t}. \quad (19)$$

Then,

$$\begin{aligned}
& N_{1,0,t} E(\text{DID}_{+,t} | \mathbf{D}) \\
&= \sum_{g:D_{g,t}=1, D_{g,t-1}=0} N_{g,t} E(\Delta_{g,t} | \mathbf{D}) + \sum_{g:D_{g,t}=1, D_{g,t-1}=0} N_{g,t} E(Y_{g,t}(0) - Y_{g,t-1}(0) | \mathbf{D}) \\
&\quad - \frac{N_{1,0,t}}{N_{0,0,t}} \sum_{g:D_{g,t}=D_{g,t-1}=0} N_{g,t} E(Y_{g,t}(0) - Y_{g,t-1}(0) | \mathbf{D}) \\
&= \sum_{g:D_{g,t}=1, D_{g,t-1}=0} N_{g,t} E(\Delta_{g,t} | \mathbf{D}) + \psi_{0,t} \left(\sum_{g:D_{g,t}=1, D_{g,t-1}=0} N_{g,t} - \frac{N_{1,0,t}}{N_{0,0,t}} \sum_{g:D_{g,t}=D_{g,t-1}=0} N_{g,t} \right) \\
&= \sum_{g:D_{g,t}=1, D_{g,t-1}=0} N_{g,t} E(\Delta_{g,t} | \mathbf{D}). \tag{20}
\end{aligned}$$

The first equality follows by (18), the second by (19), and the third after some algebra. If there is no g such that $D_{g,t-1} = 0$ and $D_{g,t} = 1$, (20) still holds, as $\text{DID}_{+,t} = 0$ in this case.

A similar reasoning yields

$$N_{0,1,t} E(\text{DID}_{-,t} | \mathbf{D}) = \sum_{g:D_{g,t}=0, D_{g,t-1}=1} N_{g,t} E(\Delta_{g,t} | \mathbf{D}) \tag{21}$$

Plugging (20) and (21) into (17) yields

$$\begin{aligned}
E(\text{DID}_M) &= \sum_{t=2}^T E \left(E \left(\frac{1}{N_S} \left(\sum_{g:D_{g,t}=1, D_{g,t-1}=0} N_{g,t} \Delta_{g,t} + \sum_{g:D_{g,t}=0, D_{g,t-1}=1} N_{g,t} \Delta_{g,t} \right) \middle| \mathbf{D} \right) \right) \\
&= \delta^S.
\end{aligned}$$

Proof of Theorem 4

First, as with DID_M , we have

$$E(\text{DID}_M^{\text{pl}}) = \sum_{t=3}^T E \left(\left(\frac{N_{1,0,0,t}}{N_S^{\text{pl}}} E(\text{DID}_{+,t}^{\text{pl}} | \mathbf{D}) + \frac{N_{0,1,1,t}}{N_S^{\text{pl}}} E(\text{DID}_{-,t}^{\text{pl}} | \mathbf{D}) \right) \right). \tag{22}$$

Let t be greater than 3, and let us for now focus on the case where there exists at least one g_1 such that $D_{g_1,t-2} = D_{g_1,t-1} = 0$ and $D_{g_1,t} = 1$. Then Assumption 13 ensures that there is a least

another group g_2 such that $D_{g_2,t-2} = D_{g_2,t-1} = D_{g_2,t} = 0$. Then,

$$\begin{aligned}
& N_{1,0,0,t} E \left(\text{DID}_{+,t}^{\text{pl}} \middle| \mathbf{D} \right) \\
&= \sum_{g: D_{g,t}=1, D_{g,t-1}=D_{g,t-2}=0} N_{g,t} E (Y_{g,t-1}(0) - Y_{g,t-2}(0) | \mathbf{D}) \\
&\quad - \frac{N_{1,0,0,t}}{N_{0,0,0,t}} \sum_{g: D_{g,t}=D_{g,t-1}=D_{g,t-2}=0} N_{g,t} E (Y_{g,t-1}(0) - Y_{g,t-2}(0) | \mathbf{D}) \\
&= \psi_{0,t-1} \left(\sum_{g: D_{g,t}=1, D_{g,t-1}=D_{g,t-2}=0} N_{g,t} - \frac{N_{1,0,0,t}}{N_{0,0,0,t}} \sum_{g: D_{g,t}=D_{g,t-1}=D_{g,t-2}=0} N_{g,t} \right) \\
&= 0.
\end{aligned} \tag{23}$$

The second equality follows by (19), and the third follows after some algebra. If there exists no g such that $D_{g,t-2} = D_{g,t-1} = 0$ and $D_{g,t} = 1$, (23) still holds, as $\text{DID}_{+,t}^{\text{pl}} = 0$ in this case.

A similar reasoning yields

$$N_{0,1,1,t} E \left(\text{DID}_{-,t}^{\text{pl}} \middle| \mathbf{D} \right) = 0 \tag{24}$$

The result follows after plugging (23) and (24) into (22).